

Muhammad Hammad Bashir

Machine Learning Engineer — Computer Vision — Generative AI

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PROFESSIONAL SUMMARY

Data Scientist and Machine Learning Engineer with 5+ years of experience across Computer Vision, deep learning, Generative AI, AI Agents, workflow automation, and multimodal AI, delivering intelligent solutions for sports analytics, advertising and marketing, chatbots, agriculture, and enterprise applications. Specialized in end-to-end video intelligence pipelines involving object detection, tracking, pose estimation, and event recognition using YOLO, MediaPipe, ByteTrack, and deep learning frameworks. Experienced in developing LLM-powered applications, Retrieval-Augmented Generation (RAG) systems, AI agent workflows, and intelligent automation solutions. Proven track record in building scalable ML systems, real-time inference pipelines, multi-camera streaming analytics, production-grade MLOps deployments, and cloud-native AI applications.

PROFESSIONAL EXPERIENCE

Sportunity / Sports Analytics Platform

Computer Vision Engineer (Sports AI)

June 2025 – Present

- Built a real-time sports computer vision system for soccer, tennis, and volleyball using live RTSP (HIKvision) camera feeds.
- Implemented **YOLOv8m + ByteTrack + TrackNet** for player, ball detection, and multi-object tracking under low-latency constraints.
- Developed event detection pipelines including ball bounce detection, trajectory interpolation, speed/acceleration estimation, and shot classification.
- Designed multi-camera synchronization system for cross-angle tracking, enabling accurate match reconstruction and highlight generation.
- Integrated **MediaPipe** pose estimation for biomechanics analysis (jump, sprint, movement efficiency).
- Optimized inference pipeline using **ONNX + TensorRT** for edge and GPU deployment.
- Built automated AI coaching system generating real-time analytics dashboards for coaches and athletes.

Markify / CreativeScore AI

London, UK

AI Consultant (Computer Vision & Generative AI)

June 2023 – June 2025

- Developed neuromarketing AI systems analyzing ad effectiveness using computer vision and multi-modal signals (image, video, audio, and text).
- Built attention prediction and saliency mapping models to identify high-impact visual regions in advertisements.
- Designed emotion recognition pipelines for viewer reaction analysis using deep CNNs and transformer-based encoders.
- Implemented **Pix2PixHD**, GAN-based augmentation, and diffusion-based creative optimization for ad generation.
- Built end-to-end **FastAPI + AWS + Azure** CI/CD pipelines for scalable deployment.
- Delivered creative intelligence platform (creativescore.ai) for optimizing marketing performance and engagement.

LUMS / AI & Big Data Centre

Pakistan

Research Assistant (AI & Computer Vision)

Sept 2021 – June 2023

- Developed ML-based livestock health monitoring systems using image classification and sensor data.
- Worked on flood and drought prediction using **LSTM** and **RNN**-based time-series models.
- Built agricultural AI systems for crop monitoring and yield prediction using satellite imagery.
- Implemented classical ML pipelines using **Scikit-learn**, XGBoost, Flask, and SQL-based deployments.

Freelance ML Engineer

Upwork & Fiverr

Computer Vision & ML Consultant

Aug 2021 – Jul 2023

- Delivered end-to-end ML and computer vision solutions for global clients in sports analytics, document AI, and geospatial intelligence.

SKILLS

Computer Vision:	YOLOv8, OpenCV, MediaPipe, ByteTrack, DeepSORT, TrackNet, SAM, CLIP
Video Analytics:	RTSP pipelines, multi-camera sync, event detection, tracking systems
Deep Learning:	PyTorch, TensorFlow, CNNs, Transformers, GANs, diffusion models
MLOps:	Docker, CI/CD, Azure DevOps, MLflow, DVC
Deployment:	FastAPI, AWS, Azure, ONNX, TensorRT, serverless GPU systems
Geospatial AI:	QGIS, GDAL, rasterio, satellite image analysis
LLMs & Multimodal AI:	GPT, Claude, LLaMA, LangChain, CrewAI

EDUCATION

Namal University	Mianwali, Pakistan
BSc in Computer Science [Specialisation in Machine Learning]; CGPA: 3.46/4.0	2017 – 2021

SELECTED PROJECTS

- ““ **Real-Time Multi-Camera Sports Vision System:**
- Built a production-grade sports AI system using live RTSP feeds for real-time match analysis.
 - Integrated **YOLOv8**, **ByteTrack**, and **TrackNet** for detection and tracking of players and ball.
 - Developed ball interpolation, bounce detection, and event recognition modules.
 - Used **MediaPipe** for pose-based biomechanics; enabled multi-camera synchronization for highlight generation and AI-based coaching feedback.
- MoversAI – Intelligent Inventory Quote Generation System:**
- Developed an AI-powered web platform allowing users to capture room-by-room images for automated moving quotes.
 - Used object detection and scene understanding to identify items, estimate volume, and generate pricing.
 - Integrated human-in-the-loop verification for accuracy and production reliability.
- Drone & Satellite Vision Intelligence (TMGDronity):**
- Worked on multimodal geospatial AI system combining satellite, drone, RGB, multispectral, and hyperspectral imagery.
 - Implemented geospatial alignment using **QGIS**, **GDAL**, and **rasterio**.
 - Built **YOLO**-based object detection and **UNet** segmentation models for vegetation, infrastructure, and underground object inference.
 - Included greenery estimation and plant counting pipelines.
- DoctorAI – Fall Detection & Health Monitoring System:**
- Developed AI-assisted healthcare monitoring system combining computer vision and LLM-based reasoning.
 - Implemented fall detection using pose estimation and anomaly detection pipelines.
 - Integrated multimodal signals for patient activity monitoring and alert generation.
- Atlas Knowledge Assistant – Enterprise AI Knowledge Automation:**
- Built a production-grade AI knowledge assistant for engineering organizations using **RAG**, specialized agents, and grounded document retrieval.
 - Implemented **Claude/OpenAI** model routing, confidence-based escalation, citation-backed responses, and per-workflow cost tracking.
 - Integrated **ClickUp** automation and **MCP** tools for task processing, SOP resolution, comments, and workflow updates.
 - Developed a **Streamlit** interface for knowledge search, task execution, and usage analytics.
- AI Receptionist – Voice-Powered Appointment System:**
- Built a production-ready voice AI system for automated hospital appointment scheduling, cancellation, and schedule lookup.
 - Integrated **VAPI** for real-time speech processing, LLM function calling, and voice interactions.
 - Developed **FastAPI** REST endpoints with **SQLAlchemy/SQLite** for appointment management and data persistence.
 - Built a **Streamlit** administration dashboard for hospital staff to view and manage appointment schedules.

CERTIFICATIONS

- Completed the **AWS Machine Learning Foundations** course in the Udacity Scholarship Program.
- Earned the **Deep Learning Specialization** certification from **DeepLearning.AI** (Coursera).
- Completed the **Machine Learning Engineering for Production (MLOps)** Specialization by **DeepLearning.AI** on Coursera.
- Completed the **Machine Learning Specialization** by **DeepLearning.AI** and Stanford Online (Coursera).
- Completed the **Computer Vision** course on Coursera.